DATA ITEM DESCRIPTION
Title: JOINT PRECISION APPROACH AND LANDING SYSTEM (JPALS) CERTIFICATION PLAN

Number: DI-MISC-82124
AMSC Number: N9806
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20170608
Limitation: None
GIDEP Applicable: N/A
Project Number: MISC-2017-008

Use/Relationship: The Joint Precision Approach and Landing System (JPALS) Certification Plan describes the Contractor's plan for delivery of Guidance, Navigation, and Control (GN&C) models, algorithms, and certification data artifacts in support of Government JPALS Certification activities.

This Data Item Description (DID) contains the format, content, and intended use information for the JPALS Certification Plan.

1. **Reference Documents:** The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the contract.

2. **Format:** Contractor's format is acceptable. If the models have been developed in a commonly available design and analysis tool (such as Matlab and Simulink) then delivery of the models in such format is acceptable.

3. **Content:** The JPALS Certification Plan shall contain the following information:

a. Airborne GN&C Models.
    1. GN&C system models and algorithms, including the inner loop and outer loop control laws.
    2. Gains, limits, and filter values needed for the GN&C models.
b. Flight Test Activities
    1. Description of how flight test activities will support the Government's JPALS Certification efforts.
c. Real-Time Certification Data
    1. Inner loop control law inputs
    2. Vertical displacement of the touchdown point, including the un-stabilized touchdown point and the touchdown point with Deck Motion Compensation (DMC) commands.
    3. Lateral displacement of the touchdown point, including the un-stabilized touchdown point and the touchdown point with DMC commands (if applicable).
    4. Description of how the real-time certification data is provided and the path from the sensor/system generating the data to the production aircraft data bus, in-flight data recorder, and telemetry system.
d. Certification Data for Post Processing
    1. Inner loop control law inputs and outputs.
    2. Outer loop control law inputs and outputs.

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3. Guidance algorithm inputs and outputs.
4. Navigation algorithm inputs and outputs.
5. Sensor outputs from systems that feed guidance and control (e.g. Embedded Global Positioning System (GPS) Inertial Navigation System (INS) (EGI), Air Data System).
6. Independent navigation reference source outputs.
7. Engine Throttle and revolutions per minute (RPM).
8. Control surface positions.
9. Stabilized and un-stabilized relative position vector.
10. Vertical displacement of the touchdown point, including the un-stabilized touchdown point and the touchdown point with DMC commands.
11. Lateral displacement of the touchdown point, including the un-stabilized touchdown point and the touchdown point with DMC commands (if applicable).
12. Description of how the data for post processing is provided and the path from the sensor/system generating the data to the inflight data recorder.

e. Appendices: As appropriate or required.

End of DI-MISC-82124

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